

Roll No. :

MID-SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: **B.Tech (CSE)**

Semester: III

Name of the Course: Mathematical Foundations for Artificial Intelligence

Course Code: TCS343

Time: 90 Minutes

Maximum Marks: 50

Note:

- i. Answer all the questions by choosing any one of the sub-questions
- ii. Each question carries 10 marks

Q1	(10 Marks)	
(a)	Explain Gaussian elimination and Gauss-Jordan elimination methods for solving systems of linear equations. Compare their computational efficiency with examples.	CO1
(b)	A company manufactures two products P and Q. Each unit of P requires 2 hours of machining and 1 hour of assembly, while each unit of Q requires 1 hour of machining and 3 hours of assembly. If the total available hours are 100 for machining and 90 for assembly, formulate the system of equations and solve using matrix inversion method.	
Q2	(10 Marks)	
(a)	Define linear independence of vectors. Determine whether the set $\{(1,2,3), (2,4,6), (3,6,9)\}$ in R^3 is linearly independent. Justify your answer.	CO2
	OR	
(b)	Consider the matrix A. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 6 & 8 \end{bmatrix}$ i. Find the rank of A. ii. Identify a basis for its row space.	
Q3	(10 Marks)	
(a)	Explain the concept of linear mapping. For the mapping $T: R^2 \rightarrow R^2$ defined by $T(x,y) = (2x + y, x - y)$, find the matrix representation of T.	CO2
	OR	
(b)	Define an affine space. Consider the affine space given by the equation $3x - 2y + z = 5$. Find one point on the affine space and give the direction vectors spanning the associated subspace.	

Q4	(10 Marks)	
(a)	Define orthogonal projection. Find the orthogonal projection of vector $x = (2,3,4)$ onto the subspace spanned by $(1,1,0)$.	CO3
	OR	
(b)	For vectors $u = (1,2,1)$, $v = (2,1,0)$, and $w = (1,1,1)$, check whether w lies in the orthogonal complement of $\text{span}\{u,v\}$.	
Q5	(10 Marks)	
(a)	Define an orthonormal basis. Apply the Gram-Schmidt process to obtain an orthonormal basis from $\{(1,1,0), (1,0,1)\}$.	CO2
	OR	
(b)	Discuss the concept of orthogonal projection in machine learning feature reduction. Illustrate with an example.	CO4